## Question Paper Code: 25901193

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM
# MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
# Fourth SEMESTER
# Discipline Specific Elective (DSE) - MG4DSECSC201, MG4DSEECC201 –

# MOBILE APP DEVELOPMENT

**Duration: 1 Hour 30 minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)***

**ANSWER KEY**

## Part A

Very Short Answer Questions
Answer all questions
Each question carries **2** marks

1. Differentiate between Android SDK and JDK. [U] / [CO1]
   - Android SDK (Software Development Kit):
     Provides tools, libraries, and APIs specifically for developing Android applications.
   - JDK (Java Development Kit):
     Provides tools to develop Java programs (compiler, runtime environment).

   SDK is Android-specific, while JDK is general for Java development.
2. Classify different types of layouts used in Android. [U] / [CO2]
   - LinearLayout – Arranges elements in a row or column
   - RelativeLayout – Positions elements relative to each other
   - ConstraintLayout – Flexible layout with constraints
   - FrameLayout – Displays a single view or overlays views
   - TableLayout – Organizes elements in rows and columns
3. What is the purpose of Sms Manager in Android? [K] / [CO3]
   - Used to send SMS messages programmatically
   - Supports text messages, multipart messages, and data messages
4. State the primary benefit of the "Single Codebase" approach. [U] / [CO1]
   - Allows developers to write code once and run on multiple platforms
   - Reduces development time and maintenance effort
5. Recall the purpose of Radio Group. [K] / [CO2]
   - Groups multiple RadioButtons together
   - Ensures only one option can be selected at a time

**[2x5 = 10]**

## Part B

Short Answer Questions
Answer any **5** questions
Each question carries **5** marks

6. Describe the role of the Application Framework in Android. [U] / [CO1]
   - Provides high-level services to application developers
   - Acts as a bridge between applications and hardware/system
   - Offers reusable components like:
     - Activity Manager (manages app lifecycle)
     - Window Manager (handles UI)
     - Content Providers (data sharing)
     - Resource Manager (access to resources)
   - Enables code reuse and faster development
   - Ensures consistency and standardization across apps
7. Discuss different types of menus in Android. [U] / [CO2]

- Options Menu
  - Main menu of the app (appears in app bar)
- Context Menu
  - Appears on long press of a view
- Popup Menu
  - Appears anchored to a specific view
- SubMenu
  - Nested menu inside another menu
- Contextual Action Mode (CAB)
  - Provides actions for selected items

8. Explain the role of HTTP and JSON in Android web service communication. [U] / [CO3]

- HTTP (HyperText Transfer Protocol):
  - Used for communication between client (Android app) and server
  - Supports request methods like GET, POST, PUT, DELETE
- JSON (JavaScript Object Notation):
  - Lightweight data format for data exchange
  - Easy to parse and generate
- Combined Role:
  - HTTP sends requests → server responds with JSON data
  - Enables data transfer, API communication, and dynamic content loading

9. Describe the 'Testing' phase and the various types of testing involved. [K] / [CO1]

- Testing Phase:
  - Ensures application is free from errors and meets requirements
  - Performed before deployment
- Types of Testing:
  - Unit Testing – Tests individual components
  - Integration Testing – Tests combined modules
  - System Testing – Tests entire application
  - User Acceptance Testing (UAT) – Validates user requirements
  - Performance Testing – Checks speed and responsiveness

10. Describe different types of fragments in Android. [U] / [CO2]

- ListFragment – Displays a list of items
- DialogFragment – Shows dialog windows
- PreferenceFragment – Used for settings UI
- WebViewFragment – Displays web content
- Custom Fragments – User-defined fragments

11. Differentiate between Content Provider and SQLite database. [U] / [CO3]

| Aspect | Content Provider | SQLite Database |
|---|---|---|
| Purpose | Data sharing between apps | Local data storage |
| Access | Accessible by multiple apps | Restricted to one app |
| Interface | Uses URI (Uniform Resource Identifier) | Uses SQL queries |
| Security | Controlled access permissions | Internal app use |
| Usage | Inter-app communication | Data storage within app |

12. Discuss the concept and importance of Fragments in Android applications. [U] / [CO2]

- Concept:
  - Fragment is a reusable UI component within an activity
  - Has its own lifecycle and layout
- Importance:
  - Enables modular UI design
  - Supports multi-pane layouts (tablets, large screens)
  - Improves code reusability
  - Allows dynamic UI changes at runtime
  - Enhances user experience and flexibility

**[5x5 = 25]**

## Part C

Essay Questions
Answer any **1** questions
Each question carries **15** marks

13. Describe the complete procedure for publishing an Android application. [U] / [CO1]

1. Prepare the Application
   - Ensure the app is fully tested and bug-free
   - Remove debug code and logs
   - Optimize performance and UI
2. Generate Signed APK / App Bundle
   - Create a keystore file
   - Sign the app using a private key
   - Generate .APK or .AAB (Android App Bundle)
3. Configure App for Release
   - Set version code and version name
   - Disable debugging (android:debuggable = false)
   - Optimize resources and enable ProGuard (optional)
4. Create Developer Account
   - Register on Google Play Console
   - Pay one-time registration fee
5. Prepare Store Listing
   - Add:
     - App name and description
     - Screenshots and icons
     - Category and tags
     - Privacy policy
6. Upload Application
   - Upload signed APK/AAB file
   - Provide content rating and target audience
7. Testing Before Release
   - Use:
     - Internal testing
     - Closed testing
     - Open testing
8. Publish the App
   - Submit for review by Google
   - After approval, app becomes available on Play Store
9. Post-Release Maintenance
   - Monitor feedback and reviews
   - Fix bugs and release updates
   - Add new features

Publishing involves preparation, signing, testing, and deployment, ensuring quality and security.

14. Demonstrate event handling and UI notifications in Android with suitable examples. [A] / [CO2]

A. Event Handling in Android

Definition:

- Event handling is the process of responding to user actions such as clicks, touches, or key presses.

Common Events:

- Button click
- Touch event
- Key press

Example (Button Click Event):

```
Button btn = findViewById(R.id.button);
btn.setOnClickListener(new View.OnClickListener() {
   @Override
```

```
    public void onClick(View v) {
        // Action to perform
        Toast.makeText(getApplicationContext(), "Button Clicked",
Toast.LENGTH_SHORT).show();
    }
});
```

B. UI Notifications in Android

Definition:

- UI notifications are used to inform users about events or updates.

Types of Notifications:

1. Toast
    - Small popup message
    - Example:

```
Toast.makeText(this, "Hello User", Toast.LENGTH_SHORT).show();
```

2. Snackbar
    - Appears at bottom with optional action

```
Snackbar.make(view, "Message", Snackbar.LENGTH_LONG).show();
```

3. Dialog Box
    - Alerts user with options

```
AlertDialog.Builder builder = new AlertDialog.Builder(this);
builder.setMessage("Are you sure?")
       .setPositiveButton("Yes", null)
       .setNegativeButton("No", null)
       .show();
```

4. Notification (Status Bar)
    - Shown outside the app
    - Uses NotificationManager

**[15x1 = 15]**